

5. System Response and LTI Properties

Experiment 5.1

Impulse Response of the System

Program:

```
% Impulse Response of the System

clc;

clear;

close all;

% Sample index

n = 0:10;

% Unit impulse input

x = [1 zeros(1,10)];

% System:  $y(n)=x(n)+0.5x(n-1)$ 

b = [1 0.5];

a = [1];

% Impulse response

h = filter(b,a,x);

% Display result

disp('Impulse Response h(n):');

disp(h);

% Plot

figure;

stem(n,h,'filled');

grid on;

title('Impulse Response of the System');

xlabel('n');

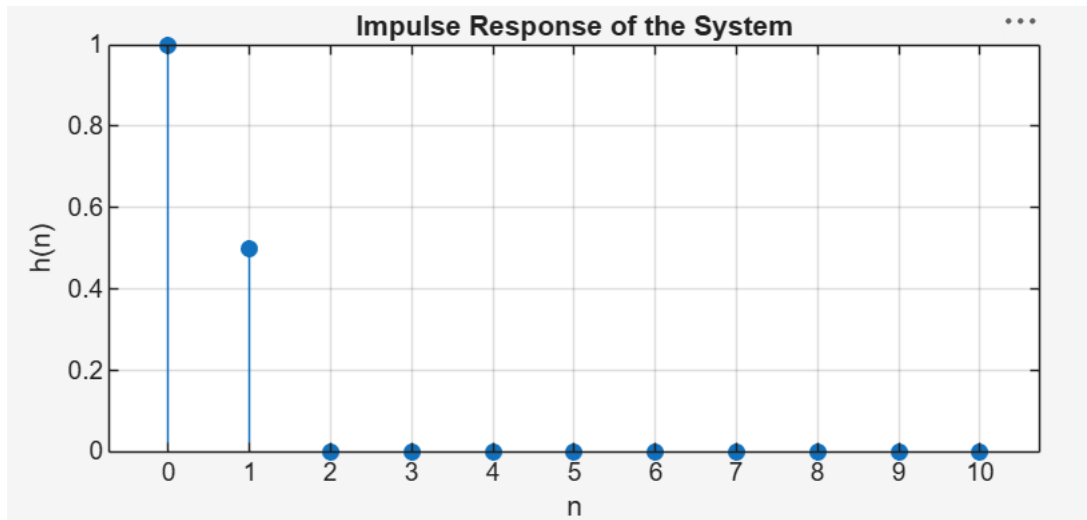
ylabel('h(n)');
```

Command Window

Impulse Response $h(n)$:

1.0000 0.5000 0 0 0 0 0 0 0 0

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Experiment 5.2

Step Response of the System

Program:

```
% Step Response of the System
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Sample index
```

```
n = 0:10;
```

```
% Unit step input
```

```
x = ones(1,length(n));
```

```
% System:  $y(n)=x(n)+0.5x(n-1)$ 
```

```
b = [1 0.5];  
a = [1];  
% Step response  
y = filter(b,a,x);  
% Display result  
disp('Step Response y(n):');  
disp(y);  
% Plot  
figure;  
subplot(2,1,1);  
stem(n,x,'filled');  
grid on;  
title('Unit Step Input u(n)');  
xlabel('n');  
ylabel('Amplitude');  
subplot(2,1,2);  
stem(n,y,'filled');  
grid on;  
title('Step Response of the System');  
xlabel('n');  
ylabel('y(n)');
```

Command Window

Step Response $y(n)$:

1.0000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
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